

# Minsoo Kim



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## Summary

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I design, build and operate an MLOps platform on on-prem Kubernetes, alone. Over six months at HUIINNO I codified the entire path — ingestion → lakehouse → transformation → lineage → distributed training → experiment / model registry — in **Pulumi (Python IaC)**, as the **sole owner of cluster construction** on a two-person platform team. I prefer removing the human bottleneck to adding tools, and I make decisions by measurement. Before this, I started as an iOS developer, moved through backend, frontend and cloud data pipelines, and led a data engineering team.

## Key Results (measured 2026-09)

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**Authored the majority of all 12 Pulumi infrastructure stacks** — 93% of stack-code changes (6 stacks above 96%). 92% of the repository's 1,204 commits and 678 merged PRs.

**10 nodes / 235 running pods / 23 internal services** in production. SSO across all 23 with Dex (GitHub OIDC) + oauth2-proxy, the same identity carried into Kubernetes RBAC.

**83% of the ETL repository's 1,599 commits** — 93% of the shared framework and 100% each of the CAST, MIMIC-IV and operational pipelines. Removed the single-person dependency in dataset preparation with Airflow 3.x + dbt + Apache Iceberg, lineage auto-recorded in DataHub.

**Training-data delivery 25× faster** — network and storage redesign took iperf3-measured throughput from 93.7 Mbps to 2.5 Gbps; an 8 TB transfer from 8.4 days to 7.8 hours.

**Reversed a GPU purchase decision with measurement** — 10 days of telemetry showed 5.3% utilization and zero queued jobs; the bottleneck was 1,112 GPU-hours held idle by dormant sessions (91% of held time). Redirected from purchase to reclamation.

**AI coding agents run continuously to lift one person's throughput to team scale** — backed by 719 SDK unit tests, CI gates, 23 incident write-ups and 30 PRDs.

**Earlier:** A StarRocks PoC measured analytical query time down from 7 s to 1.4 s (80%, Cellkey). Led the AWS EMR / PySpark art-market ETL pipeline as team lead (Eazel).

## What I Do Now

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Data Engineer, HUIINNO (Dec 2025 -). I operate every layer of the MLOps platform on bare-metal Kubernetes — networking, storage, catalog, lineage, orchestration, distributed training and identity — and I am attaching a reproducibility contract (version, content hash, code commit, environment lock and seed stamped onto a dataset at execution time) plus MLflow tracking and model registry so that research code can move onto KubeRay distributed training. The design started from a quantified survey of the existing research workflow, which identified the root cause as results having no coordinates. I am establishing the split where researchers own domain ETL and the platform owns execution, lineage, verification and retries.

## Stack

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**Platform:** Kubernetes (kubeadm) · Pulumi · Cilium · SeaweedFS · Dex/OIDC · Prometheus/Grafana **Data:** Airflow 3.x · dbt · Apache Iceberg · DataHub · PostgreSQL (pg\_duckdb) · ClickHouse **ML:** Ray/KubeRay · MLflow · PyTorch · DVC **Languages & other:** Python · SQL · Swift · Shell · AWS/GCP · LangChain

## Details

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Full experience and project details are in a separate document ([resume\\_english.pdf](#)). [github.com/msk-psp](https://github.com/msk-psp)  
[linkedin.com/in/minsoo-kim-167405131](https://linkedin.com/in/minsoo-kim-167405131)